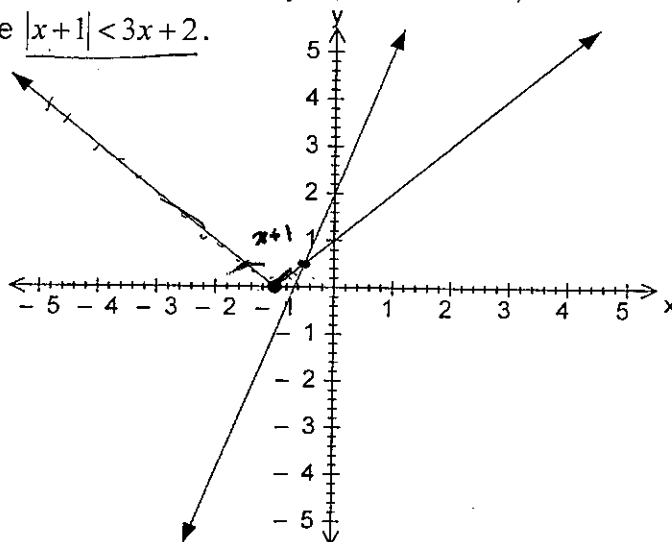


## Question 1 (14 Marks)

Marks

- (a) Solve  $x^2 - 6x + 3 = 0$  by completing the square. Give your answer in surd form. 3
- (b) Solve simultaneously  $y - 2x = 1$  and  $x^2 + y^2 = 10$  3
- (c) Solve
- (i)  $x^2 \geq 16$  1
- (ii)  $|2x - 1| < 3$  2
- (iii)  $\frac{x+3}{x-3} \geq 3$  3
- (d) The graph below shows the functions  $y = |x+1|$  and  $y = 3x+2$ . 2

Using the graph to assist you, or otherwise,  
solve  $|x+1| < 3x+2$ .



**Question 2 (16 Marks) START ON A NEW PAGE** **Marks**

- (a) Consider the function  $f(x) = \sqrt{x+1}$ . 6
- (i) State the domain and range of  $f(x)$
- (ii) Sketch the function
- (iii) Find the inverse  $f^{-1}(x)$
- (b) Sketch the graph of  $y = -(x-2)^3$  showing its key features. 2
- (c) Show whether the  $f(x) = \frac{2x}{3(1+x^2)}$  is an odd function, an even function or neither. 2
- (d) Consider the function  $f(x) = \frac{2x}{x^2 + 2x - 3}$ . 6
- (i) Find the x and y intercepts
- (ii) Find the equations of any vertical asymptotes
- (iii) Find the equation of the horizontal asymptote
- (iv) Sketch the graph of the function.

**Question 3 (23 Marks) START ON A NEW PAGE**

- (a) Find the exact value of  $\cot 120^\circ$ . 1
- \* (b) If  $\tan \theta = 2$  and  $\sin \theta < 0$ , find the exact value of  $\cos \theta$ . 2
- (c) Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ . 7
- (i)  $2 \sin \theta = -\sqrt{3}$
- (ii)  $3 \cos^2 \theta + 2 \cos \theta = 0$  (nearest minute)
- \* (iii)  $\sec \frac{\theta}{2} = -2$
- (d) Eliminate  $\theta$  from the pair of equations to find a relationship between x and y: 2
- $x = 1 + \sin \theta$
- $y = 1 + \cos \theta$
- (e) Show that  $\frac{\cot \theta}{\operatorname{cosec} \theta - 1} + \frac{\cot \theta}{\operatorname{cosec} \theta + 1} = 2 \sec \theta$  3

Q3 continued on next page

## Question 3 continued

Marks

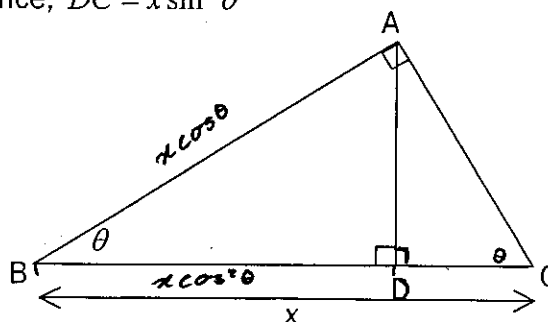
\* (f)  $\triangle ABC$  and  $\triangle ABD$  are right-angled. Show that

4

(i)  $AB = x \cos \theta$

(ii)  $BD = x \cos^2 \theta$

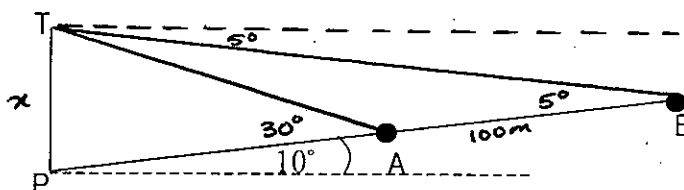
(iii) Hence,  $DC = x \sin^2 \theta$



(g) P, A, and B are points on a ramp which makes an angle of  $10^\circ$  to the horizontal as shown on the diagram below. From A and B, the angles of elevation of the top T of a flagpole at P were  $30^\circ$  and  $5^\circ$  respectively. The distance AB is 100 metres.

4

- (i) Copy and complete the diagram below.  
 (ii) Calculate the height of the tower to the nearest centimetre.



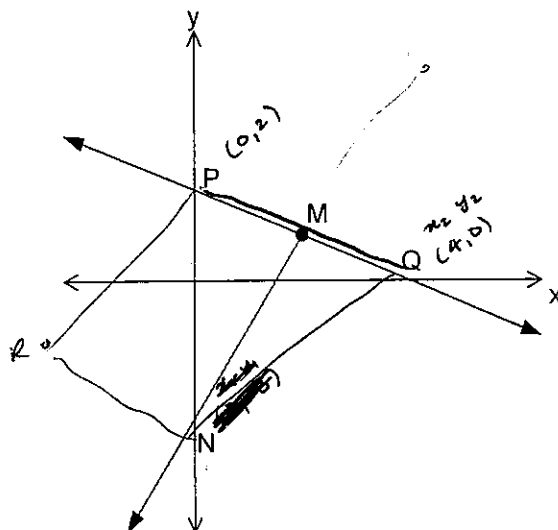
Q4 on next page

## Question 4 (16 marks) START ON A NEW PAGE

Marks

- (a) Find the coordinates of the point P that divides the interval AB, where A is (-3,5) and B is (-6,-10), externally in the ratio 2:3. 3

(b)



The diagram above shows the points  $P(x_1, y_1)$  and  $Q(x_2, y_2)$ . The point M is the midpoint of PQ. The line MN is perpendicular to PQ and meets the y-axis at N.

- (i) Show that the gradient of PQ is  $-\frac{1}{2}$ . 1
  - (ii) Find the coordinates of M. 1
  - (iii) Find the equation of the line MN. 2
  - (iv) Show that N has coordinates (0,-3). 1
  - (v) Find the distance NQ. 1
  - (vi) Find the equation of the circle with centre N and radius NQ. 2
  - (vii) Show that this circle passes through the point P. 1
  - (viii) Find the coordinates of the point R such that PRQN is a rhombus. 1
- (c) The perpendicular distance between the point (a,-2) and the line  $3y = 4x + 2$  is 8 units. Find any possible values of a. 3

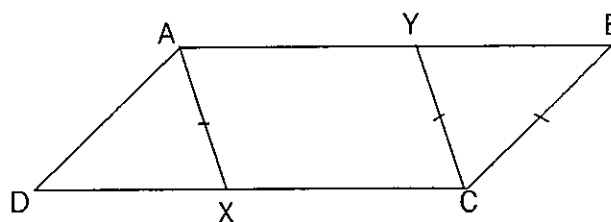
Q5 on next page

## Question 5 (16 Marks) START ON A NEW PAGE

Marks

(a)

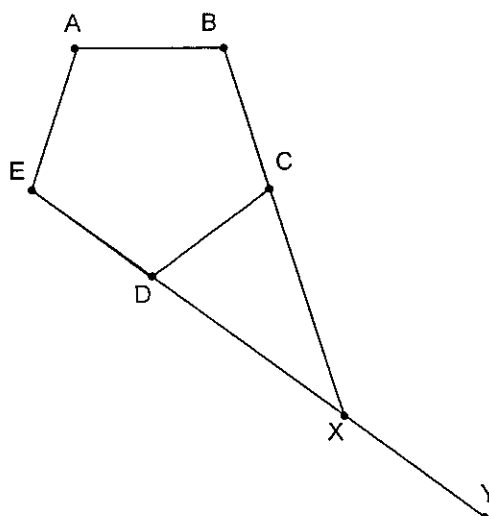
6



ABCD is a parallelogram. The point X lies on CD and the point Y lies on AB.  $AX = YC = BC$  as shown on the diagram.

- (i) Explain why  $\hat{AD}X = \hat{CB}Y$ .
- (ii) Show that  $AD = AX$
- (iii) Show that triangles ADX and CBY are congruent.
- (iv) Hence prove that AYCX is a parallelogram.

(b)



In the diagram ABCDE is a regular pentagon and BC and ED produced meet at X. The point Y lies on EDX produced. 3

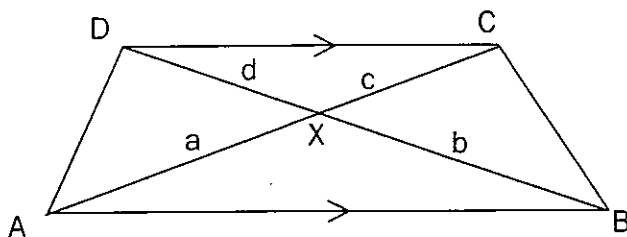
- (i) Find the size of  $\hat{BCD}$ .
- (ii) Find the size of  $\hat{CXY}$  giving reasons.

Q5 continued on next page

Question 5 continued

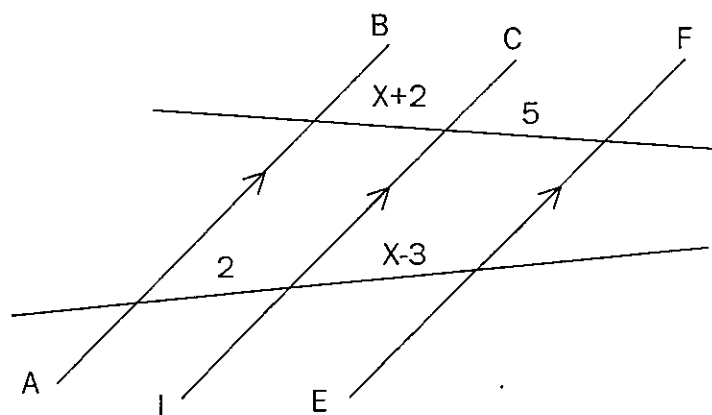
Marks

(c) 4



- (i) Prove that  $\triangle DXC \cong \triangle BXA$ .
- (ii) Hence, prove that  $\triangle DAX$  and  $\triangle CBX$  have the same area.

(d) 3



Given that  $AB \parallel CD \parallel EF$ , find the value of  $x$  giving reasons.

END OF PAPER

Q1

a)  $x^2 - 6x = -3$   
 $x^2 - 6x + \left(\frac{-6}{2}\right)^2 = -3 + \left(\frac{-6}{2}\right)^2$   
 $x^2 - 6x + 9 = -3 + 9$   
 $(x-3)^2 = 6$   
 $x-3 = \pm\sqrt{6}$   
 $x = 3 \pm \sqrt{6}$

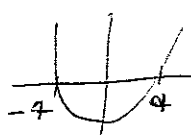
b)  $x^2 + y^2 = 10$  ①  
 $y - 2x = 1$  ②  
 From ②  $y = 2x + 1$   
 sub  $y = 2x + 1$  in ①  
 $x^2 + (2x+1)^2 = 10$   
 $x^2 + 4x^2 + 4x + 1 = 10$   
 $5x^2 + 4x - 9 = 0$   
 $(5x+9)(x-1) = 0$   
 $x = -9/5$  or  $1$

$x = -9/5$   $y = 2(-9/5) + 1$   
 $= -13/5$

$(-9/5, -13/5)$

$x = 1$   $y = 3$   
 $(1, 3)$

c) i)  $x^2 \geq 16$

$x^2 - 16 \geq 0$  

$(x-4)(x+4) \geq 0$

$x \leq -4$  or  $x \geq 4$

ii)  $|2x-1| < 3$   
 $\pm(2x-1) < 3$   
 $2x-1 < 3$   $-(2x-1) < 3$   
 $2x < 4$   $-2x+1 < 3$   
 $x < 2$   $-2x < 2$   
 $x > -1$   
 $-1 < x < 2$

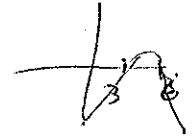
iii)  $\frac{x+3}{x-3} \geq 3$   $x \neq 3$   
 $x(x-3)^2 \times (x-3)^2$

$(x+3)(x-3) \geq 3(x-3)^2$

$(x+3)(x-3) - 3(x-3)^2 \geq 0$

$(x-3)(x+3-3(x-3)) \geq 0$

$(x-3)(-2x+12) \geq 0$

$3 \leq x \leq 6$  

but  $x \neq 3$

$\therefore 3 < x \leq 6$

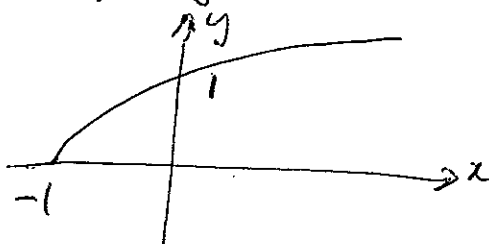
d)  $y = x+1$  ① (positive branch)  
 $y = 3x+2$  ② of  $y = |x+1|$   
 $① = ②$   
 $0 = -2x-1$   
 $x = -1/2$

$\therefore x > -1/2$

Q2

a) i) domain  $x \geq -1$ ~~range~~ range  $y \geq 0$ 

ii)



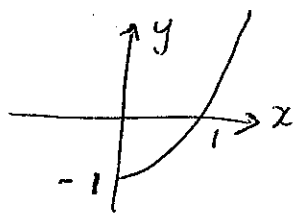
iii)

$$x = \sqrt{y+1}$$

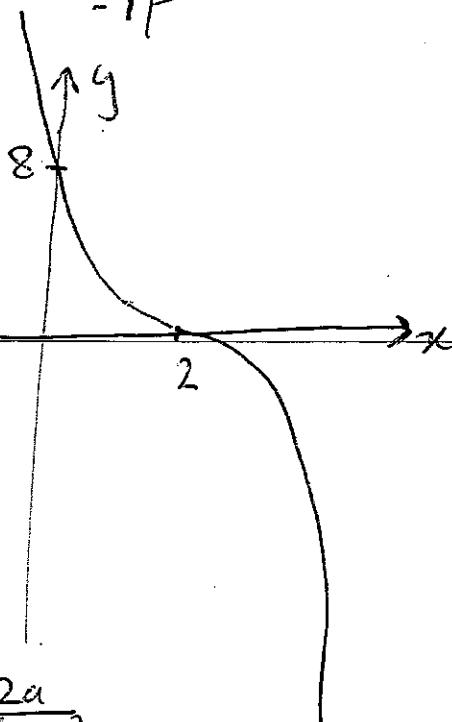
$$x^2 = y+1$$

$$y = x^2 - 1$$

$$\text{for } x \geq 0$$



b)



$$c) f(a) = \frac{2a}{3(1+a^2)}$$

$$f(-a) = \frac{2(-a)}{3(1+(-a)^2)} = \frac{-2a}{3(1+a^2)}$$

$$= -f(a)$$

 $\therefore$  ODD

$$d) f(x) = \frac{2x}{x^2+2x-3}$$

$$= \frac{2x}{(x+3)(x-1)}$$

$$i) x=0 \quad y=0 \quad (0,0)$$

 $x$  &  $y$  intercept is the origin

$$ii) x = -3 \text{ and } x = 1$$

$$iii) \lim_{x \rightarrow \infty} \frac{2x}{x^2+2x-3}$$

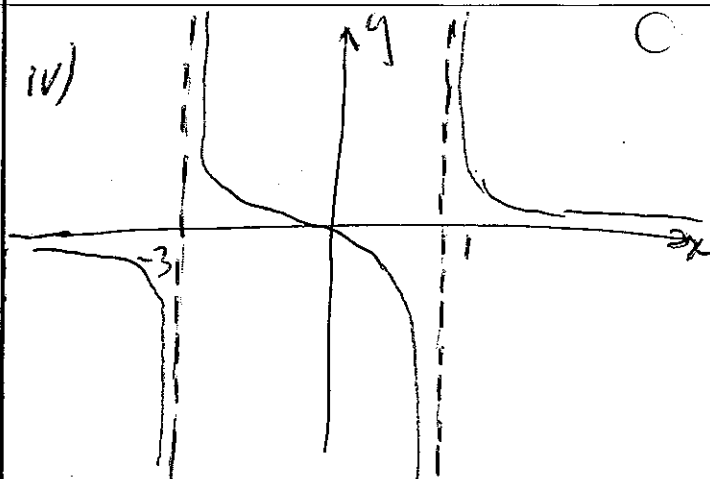
$$= \lim_{x \rightarrow \infty} \frac{\frac{2x}{x^2}}{\frac{x^2}{x^2} + \frac{2x}{x^2} - \frac{3}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2}{x}}{1 + \frac{2}{x} - \frac{3}{x^2}}$$

$$= \frac{0}{1} = 0$$

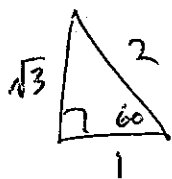
$$\therefore y = 0$$

iv)



|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| $x$ | -4  | -1  | 1/2 | 2   |
| $y$ | -16 | 0.5 | 0.2 | 0.8 |

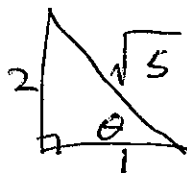
Q3 a)  $\cot 120 = \frac{-1}{\tan 60}$   
 $= \frac{-1}{\sqrt{3}}$



b)  $\tan \theta = 2$

$\sin \theta < 2$

$\cos \theta = \frac{1}{\sqrt{5}}$



c) i)  $\sin \theta = -\sqrt{3}/2$

$\theta = 180 + 60, 360 - 60$   
 $= 240, 300^\circ$

ii)  $3 \cos^2 \theta + 2 \cos \theta = 0$

$\cos \theta (3 \cos \theta + 2) = 0$

$\cos \theta = 0$  or  $\cos \theta = -\frac{2}{3}$

$\theta = 90^\circ$  or  $270^\circ$   $\theta = 180 - 48^\circ$   
 or  $180 + 48^\circ$

$\therefore \theta = 90^\circ, 270^\circ, 131^\circ 49', 228^\circ 11'$

iii)  $\sec \frac{\theta}{2} = -2$

$\cos \frac{\theta}{2} = -\frac{1}{2}$

$\frac{\theta}{2} = 180 - 60, 180 + 60$

$= 120, 240$

$\theta = 240$  or  $480$

But  $0 \leq \theta \leq 360$

$\therefore \theta = 240^\circ$

d)  $x = 1 + \sin \theta$  ①

$y = 1 + \cos \theta$  ②

From ①  $\sin \theta = x - 1$

From ②  $\cos \theta = y - 1$

$\sin^2 \theta + \cos^2 \theta = 1$

$\therefore (x-1)^2 + (y-1)^2 = 1$

e) LHS =  $\frac{\cot \theta}{\operatorname{cosec} \theta - 1} + \frac{\cot \theta}{\operatorname{cosec} \theta + 1}$

$= \frac{\cot \theta (\operatorname{cosec} \theta + 1) + \cot \theta (\operatorname{cosec} \theta - 1)}{\operatorname{cosec}^2 \theta - 1}$

$= \frac{2 \cot \theta \operatorname{cosec} \theta}{\operatorname{cosec}^2 \theta - 1}$

From  $\sin^2 \theta + \cos^2 \theta = 1$   
 $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$   
 $\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$

$= \frac{2 \cot \theta \operatorname{cosec} \theta}{\cot^2 \theta}$

$= \frac{2 \operatorname{cosec} \theta}{\cot \theta}$

$= \frac{2}{\sin \theta} \times \frac{\sin \theta}{\cos \theta}$

$= \frac{2}{\cos \theta}$

$= 2 \sec \theta = \text{RHS.}$

Q 3 continued

$$f) i) \cos \theta = \frac{AB}{x}$$

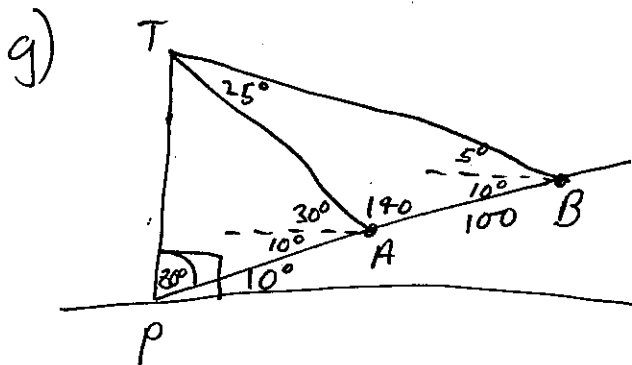
$$\therefore AB = x \cos \theta$$

$$ii) \cos \theta = \frac{BD}{AB}$$

$$\cos \theta = \frac{BD}{x \cos \theta}$$

$$BD = x \cos^2 \theta$$

$$\begin{aligned} iii) BC &= BC - BD \\ &= x - x \cos^2 \theta \\ &= x(1 - \cos^2 \theta) \\ &= x \sin^2 \theta \end{aligned}$$

In  $\triangle ATB$ 

$$\frac{100}{\sin 25^\circ} = \frac{AT}{\sin 15^\circ}$$

$$AT = \frac{100 \sin 15}{\sin 25}$$

In  $\triangle APT$ 

$$\frac{AT}{\sin 80} = \frac{TP}{\sin 40}$$

$$TP = \frac{\sin 40 \left( \frac{100 \sin 15}{\sin 25} \right)}{\sin 80}$$

$$= 39.97 \text{ m}$$

Q 4

$$a) \begin{matrix} (-3, 5) & (-6, -10) \\ & -2:3 \end{matrix}$$

$$\left( \frac{3x-3+(-2)x-6}{-2+3}, \frac{3x5+(-2)x(-10)}{-2+3} \right)$$

$$= \frac{-9+12}{1}, \frac{15+20}{1}$$

$$= (3, 35)$$

$$\begin{aligned} b) i) m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - 2}{4 - 0} \\ &= \frac{-2}{4} \\ &= -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} ii) \text{Midpt} &= \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left( \frac{0 + 4}{2}, \frac{2 + 0}{2} \right) \\ &= (2, 1) \end{aligned}$$

$$iii) m = \frac{-1}{-\frac{1}{2}} = 2 \quad (2, 1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 2)$$

$$y = 2x - 4 + 1$$

$$y = 2x - 3$$

$$iv) x = 0 \quad y = 2(0) - 3 = -3$$

$$\therefore N \text{ is } (0, -3)$$

$$v) (0, -3) (4, 0)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(0 - 4)^2 + (-3 - 0)^2}$$

$$= \sqrt{16 + 9} = 5$$

Q4 continued

$$b) (x-0)^2 + (y+3)^2 = 25$$

vi)

$$vii) x=0$$

$$(0-0)^2 + (y+3)^2 = 25$$

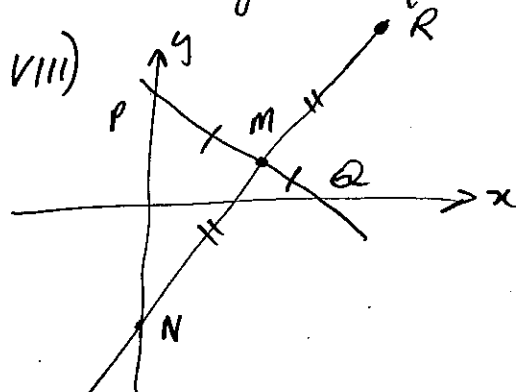
$$(y+3)^2 = 25$$

$$y+3 = \pm\sqrt{25}$$

$$y = -3 \pm 5$$

$$= 2 \text{ or } -8$$

$P(0, 2) \therefore$  circle passes through the point  $P$



$M$  is the mid point of  $NR$  if  $PRQN$  is a rhombus.

Let  $R$  be  $(x, y)$

$$\text{Then } (2, 1) = \left( \frac{x+0}{2}, \frac{y+(-3)}{2} \right)$$

$$\frac{x}{2} = 2$$

$$x = 4$$

$$\frac{y-3}{2} = 1$$

$$y-3 = 2$$

$$y = 5$$

$\therefore R$  is  $(4, 5)$

$$c) 0 = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$4x - 3y + 2 = 0$$

$$0 = 8 \quad (a, -2)$$

$$8 = \frac{|4a + -3(-2) + 2|}{\sqrt{4^2 + (-3)^2}}$$

$$8 = \frac{|4a + 8|}{5}$$

$$40 = |4a + 8|$$

$$4a + 8 = 40 \quad | \quad -(4a + 8) = 40$$

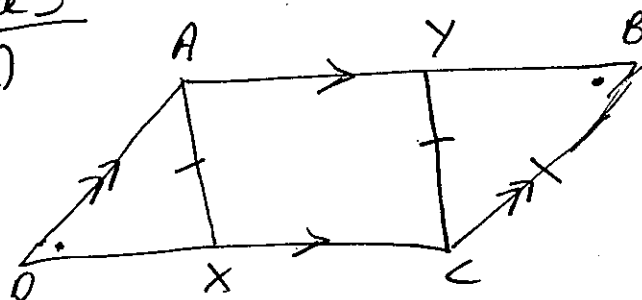
$$4a = 32 \quad | \quad -4a - 8 = 40$$

$$a = 8 \quad | \quad -4a = 48$$

$$a = -6$$

Q5

a)



i)  $\angle ADX = \angle CBY$   
(opposite angles in a parallelogram)

ii)  $AD = BC$  (opposite sides of a parallelogram)

$BC = AX$  (given)

$\therefore AD = AX$

iii) In  $\Delta s ADX$  and  $CBY$

$\angle ADX = \angle YBC$  (above)

$\angle ADX = \angle AXD$  (base angles of an isosceles  $\Delta$ )

Q5 continued

$\angle CYB = \angle CBY$  (base angles of an isosceles  $\Delta$ )

$$\therefore \angle AXD = \angle BYC$$

$$AX = CY \text{ (given)}$$

$$\therefore \Delta ADX \equiv \Delta CBY \text{ (AAS)}$$

$$\text{iv) } DX = BY \text{ (corresponding sides of congruent } \Delta\text{s)}$$

$$AB = DC \text{ (opposite sides of a parallelogram)}$$

$$XC = DC - DX$$

$$AY = AB - BY$$

$$\therefore XC = AY$$

$$AX = YC \text{ given}$$

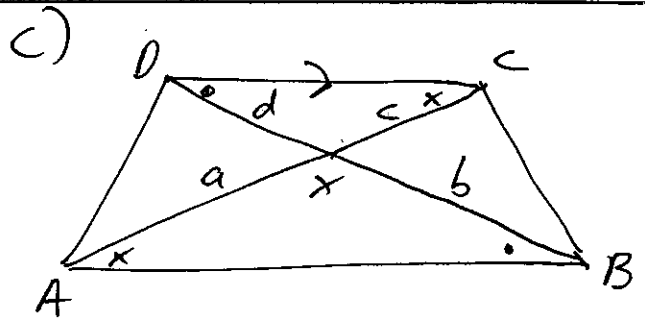
$\therefore AXCX$  is a parallelogram (opposite sides equal)

$$\begin{aligned} \text{b) } \angle BCD &= \frac{180 \times (5-2)}{5} \\ &= 108^\circ \end{aligned}$$

$$\begin{aligned} \angle XCD &= 180 - 108 \\ &= 72^\circ \text{ (supplementary } \angle\text{s)} \end{aligned}$$

$$\text{Similarly } \angle CDX = 72^\circ$$

$$\therefore \angle CXY = 72 + 72^\circ = 144^\circ \text{ (exterior angle of a triangle)}$$



i) In  $\Delta$ s  $DXC$  and  $BXA$

$$\angle CDX = \angle XBA \text{ (alternate } \angle\text{s } DC \parallel AB)$$

$$\angle DCX = \angle XAB \text{ (alternate } \angle\text{s } DC \parallel AB)$$

$$\therefore \Delta DXC \equiv \Delta BXA \text{ (equiangular)}$$

$$\text{ii) } \frac{b}{d} = \frac{a}{c} \text{ (corresponding sides in same ratio)}$$

$$\therefore bc = ad$$

$$\text{Area } \Delta DAX = \frac{1}{2} ad \sin \angle AXD$$

$$\Delta CXB = \frac{1}{2} bc \sin \angle CXB$$

$$\angle CXB = \angle AXD \text{ (vert. opp)}$$

$\therefore$  areas are equal.

$$\text{d) } \frac{2}{x-3} = \frac{x+2}{5}$$

$$10 = (x-3)(x+2)$$

$$10 = x^2 - x - 6$$

$$0 = x^2 - x - 16$$

$$x = \frac{-(-1) \pm \sqrt{1^2 - 4 \times (-16)}}{2}$$

$$= \frac{1 \pm \sqrt{1+64}}{2}$$

$$= \frac{1 \pm \sqrt{65}}{2}$$

(parallel lines cut intercepts in the same ratio)